

A prospective assay for hypothesis-space expansion in AI systems

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Abstract

Scientific discovery can require changing not only a hypothesis but the language in which hypotheses are expressed. We introduce a prospective assay for hypothesis-space expansion. A candidate must be structurally outside a frozen incumbent language, fit observations, commit to a discriminating intervention before its outcome is revealed, outperform the best incumbent predictor and survive independent falsification. Typed high-level proposals succeeded in 142/400 synthetic worlds, whereas direct, sampled, fixed-space and attribute-only alternatives succeeded in 0-1. Generic local rewrites succeeded in 400/400 known-family and 100/100 held-out worlds, but a component audit reproduced these results without model outputs. On a fixed 96-world sensitivity panel, autonomous proposals remained 0/96 after increasing Phi-4 precision or budget, substituting DeepSeek under matched or native reasoning, or allowing one validator repair; attrition remained predominantly pre-executable. DeepSeek succeeded in 3/40 worlds with a supplied representation. The assay enables auditable attribution while exposing interface dependence, saturation and limited external validity.

Introduction

Abduction selects an explanatory hypothesis, but scientific change sometimes alters the representational vocabulary in which hypotheses can be stated¹⁻³. Computational creativity and constructive-induction research formalized related distinctions between exploration within a space and transformations of that space⁴⁻⁹. Symbolic regression, program search and language-model-guided systems can discover useful equations, algorithms and hypotheses¹⁰⁻¹⁷, while multi-agent systems increasingly automate broader scientific workflows¹⁸⁻²⁰.

Recent work now explicitly targets expanding principle or model spaces. PiEvo treats discovery as optimization over an evolving principle space across four benchmarks²¹. Model Discovery Agent couples a language-model proposer to Bayesian experiment design in an open model setting spanning physics, chemistry and biology²². HypoArena evaluates prospective hypothesis discovery in 988 cases across six domains and 15 frontier models²³. EvoSCM evolves structural causal models through intervention and prospective prediction²⁴. These studies broaden the scientific tasks and models under evaluation. A complementary measurement problem remains: how can an evaluation prove that a candidate left a specified incumbent hypothesis language, charge it a prediction before seeing the answer, and separately identify which system component caused the escape?

Here we operationalize bounded hypothesis-space expansion. We freeze an incumbent representational language, generate executable candidates and require a six-gate verdict: incumbent adequacy (J0), canonical structural non-membership (J1), candidate adequacy (J2), a discriminating committed prediction (J3), prospective intervention gain (J4) and independent falsification survival (J5; Fig. 1). Candidate prose, model confidence, embedding distance and semantic novelty do not enter the verdict.

Our experiments validate the assay and reveal a component-attribution limit. Typed proposals and generic compositional search cross the frozen language reliably, whereas fixed-space alternatives do not. However, a code-path audit and inference-free replay show that C3's representations, fitted equations, structural ranking and interventions were deterministic; removing the model output leaves all 500 jump and 300 control verdicts unchanged. The central contribution is therefore the prospective assay and a transparent typed search scaffold-not evidence that a language model was necessary for the successful escapes.

Table 1 | Position relative to adjacent discovery evaluations

Evaluation	Exec	Frozen	Non-member	Prospective	Falsify	Attribution / replay	Breadth
Hypothesis Search / HypoGen	partial	no	no	no	no	no	language tasks
POPPER	partial	no	no	yes	yes	limited	literature
FunSearch	yes	skeleton	no	feedback	tests	proposer/evaluator	mathematics
PIEvo	yes	evolving	no	varies	varies	proposer/search	4 benchmarks
Model Discovery Agent	yes	open set	no	Bayesian	predictive	proposer/inference	3 sciences
HypoArena	judged	no	no	context	rubric	no	988 / 6 / 15
EvoSCM	yes	evolving	no	active	prospective	evolution/selection	causal systems
This work	yes	yes	canonical	locked	exact	factorial / exact	9 families / 2-checkpoint sensitivity

Results

A prospective criterion for hypothesis-space expansion

Figure 1 | A prospective assay for bounded representation escape

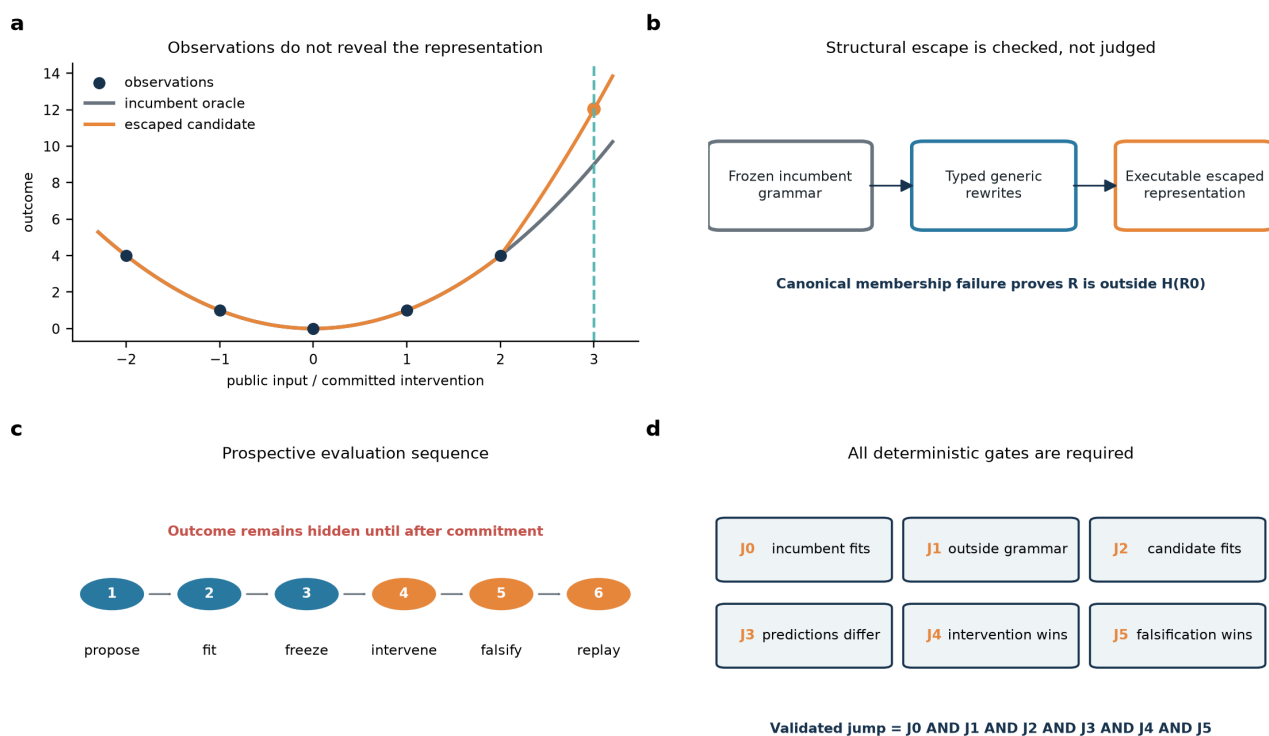


Figure 1 | The assay combines canonical structural non-membership with a prospective intervention and independent falsification. Panel a is schematic; panels b-d summarize the registered pipeline.

Each synthetic world contains observations compatible with an incumbent model, a frozen typed-graph language, hidden intervention outcomes and independent falsification cases. The best incumbent predictor is computed exactly or by bounded exhaustive search. Before any intervention outcome is revealed, a candidate representation, executable expression, selected action, prediction and split hash are committed. A validated jump is the conjunction $J0 \wedge J1 \wedge J2 \wedge J3 \wedge J4 \wedge J5$ (Fig. 1). This design turns "leaving a hypothesis space" into a replayable event rather than a judgement about wording or apparent novelty.

Typed proposals outperform fixed-space alternatives

Figure 2 | Typed proposals and their gate attrition

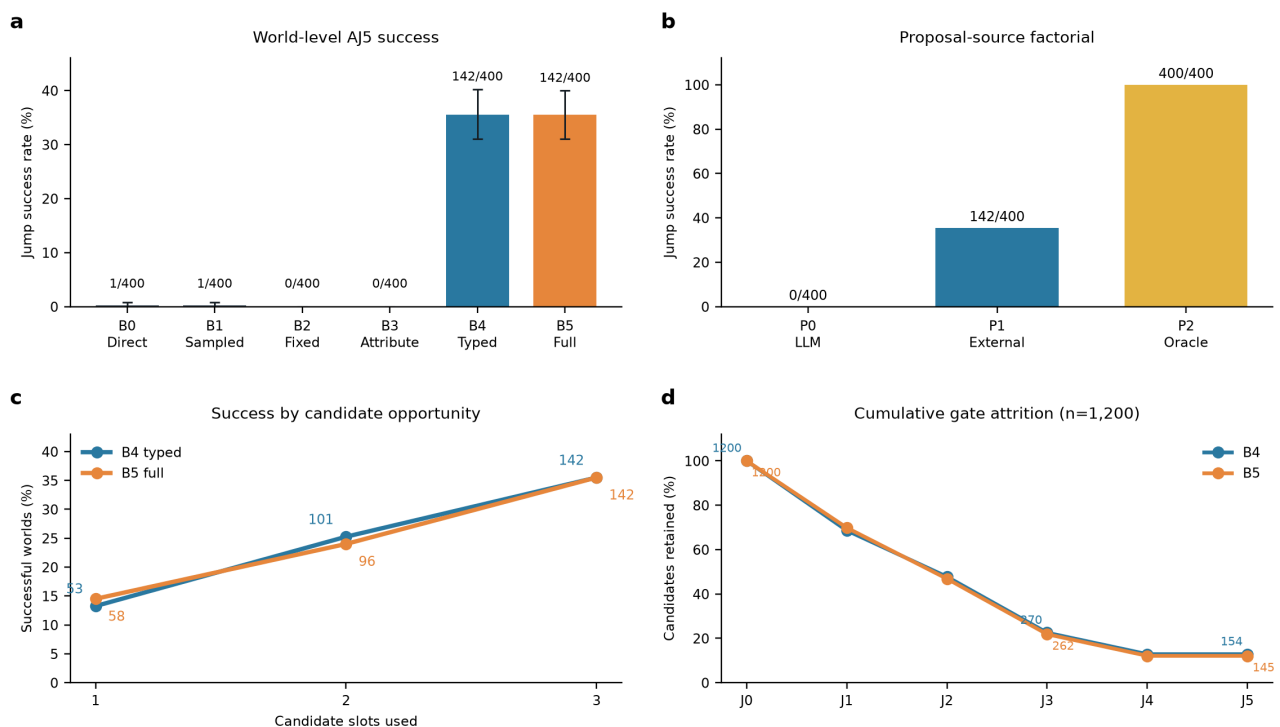


Figure 2 | AJ5 world-level jump success. Error bars are registered family-stratified bootstrap 95% intervals. Counts are successful worlds; n=400 per jump condition. All conditions recorded 0/200 false jumps.

In AJ5, direct model proposals (B0) and sampling-matched proposals (B1) each succeeded in 1 of 400 worlds. Fixed-space reasoning (B2), attribute-only mutation (B3) and value-only mutation succeeded in none. External typed representation proposals succeeded in 142 of 400 worlds in both B4 and B5 (35.5%; Fig. 2a). All eight registered comparisons of B4 or B5 against B0-B3 were positive after Holm correction (adjusted $P \leq 4.00 \times 10^{-4}$).

The proposal-source factorial held the downstream path fixed. Model-proposed representations (P0) succeeded in 0 of 400 worlds, external typed proposals (P1) in 142, and oracle representations (P2) in 400 (Fig. 2b). This shows that the tested system can execute a supplied changed representation. It does not establish that the model is needed after deterministic fitting, because the fitted expression and maximum-separation intervention were supplied to the second call and programmatically enforced.

Candidate-level gate attrition clarifies the result (Fig. 2c). In jump worlds, B4 had 1,200 candidates at J0, 823 through J1, 573 through J2, 270 through J3 and 154 through J4-J5. B5 had 1,200, 838, 562, 262 and 145, respectively. In controls, B4 and B5 still produced 112 and 110 candidates through J3, but none passed J4. The zero-control result is therefore not created by J0-J3 alone; prospective outcomes

eliminate apparently admissible escapes.

Generic rewrites compose into validated representations

Figure 3 | Generic rewrites compose into validated representations

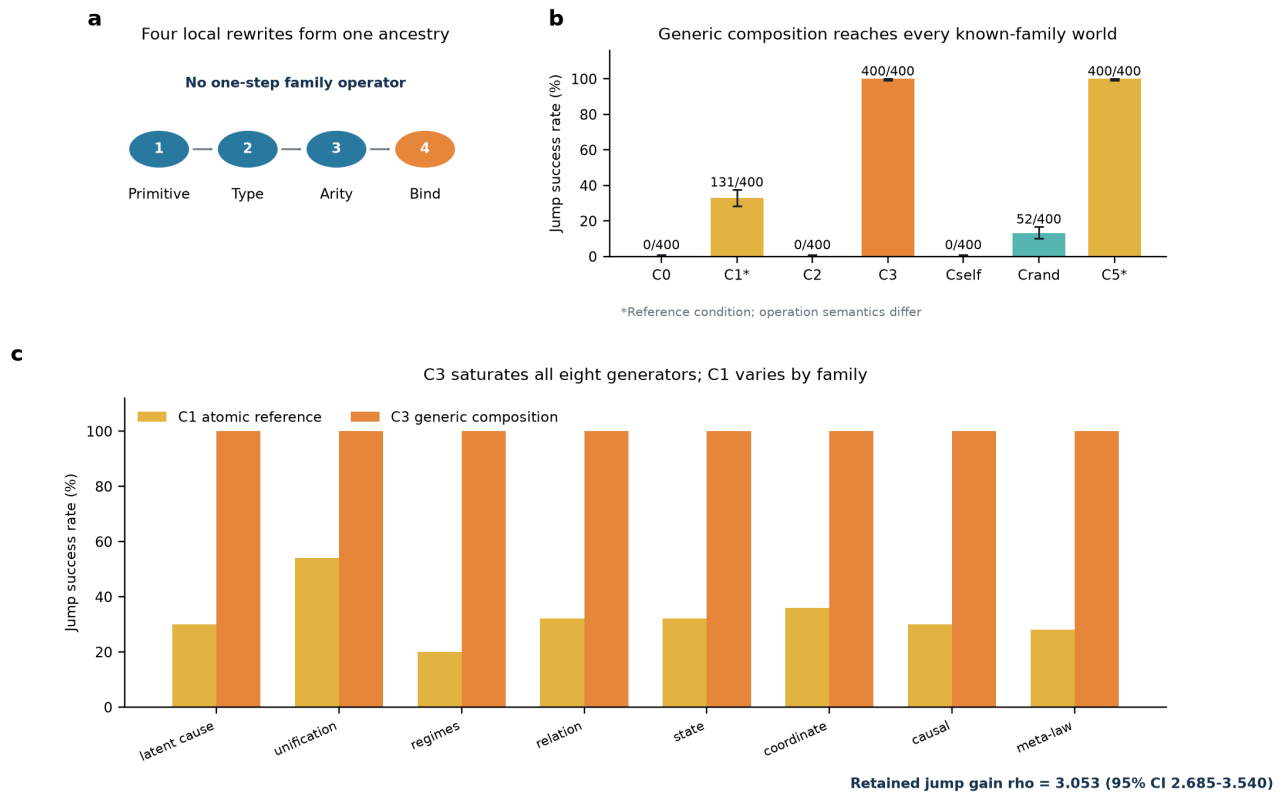


Figure 3 | CJ5 known-family reconstruction. Error bars in panel b are Wilson 95% intervals; $n=400$ worlds per condition. C1 and C5 are reference conditions with different operation semantics. Per-family panels contain 50 worlds each.

CJ5 replaced nine high-level mutations with 29 local graph and abstract-syntax-tree rewrites. Adding a node, changing its type or observability, changing arity and binding each argument were separate steps. No primitive accepted a family label, target distance or outcome. C3 deterministically traversed 48 four-step branches, ranked candidates using observational fit, structural escape and prospective prediction separation without outcomes, and retained three.

Across eight known structural families with new seeds ($n=400$ worlds), fixed-space search (C0) and 192 generic depth-one alternatives (C2) succeeded in no worlds. The frozen atomic reference (C1) succeeded in 131 (32.75%), random four-step paths (C_rand) in 52 (13.0%), and C3 in all 400 (Fig. 3). C3 exceeded C_rand by 0.87 (95% CI 0.845-0.895; Holm-adjusted $P=3.00 \times 10^{-4}$). C3 was 100% in every family; this demonstrates within-generator reliability and assay separability, not eight statistically independent replications or an unsaturated frontier benchmark.

C_self also scored 0/400, but its failure is narrower than initially anticipated. Offline reconstruction made no new model calls. The 400 known-family worlds yielded 1,200 non-empty proposal responses, each carrying 16 registered plan opportunities. The legacy parser extracted a JSON object from every response, but none had a valid outer plans list; consequently 0/19,200 plan opportunities reached execution or J1. All 1,200 responses reached the 700-token completion cap, and none was strict whole-response JSON. The historical 0/400 therefore cannot distinguish weak representation proposals from truncation and serialization failure.

Targeted sensitivities do not remove the proposal-interface bottleneck

Figure 4 | Transfer, replay and targeted model sensitivity

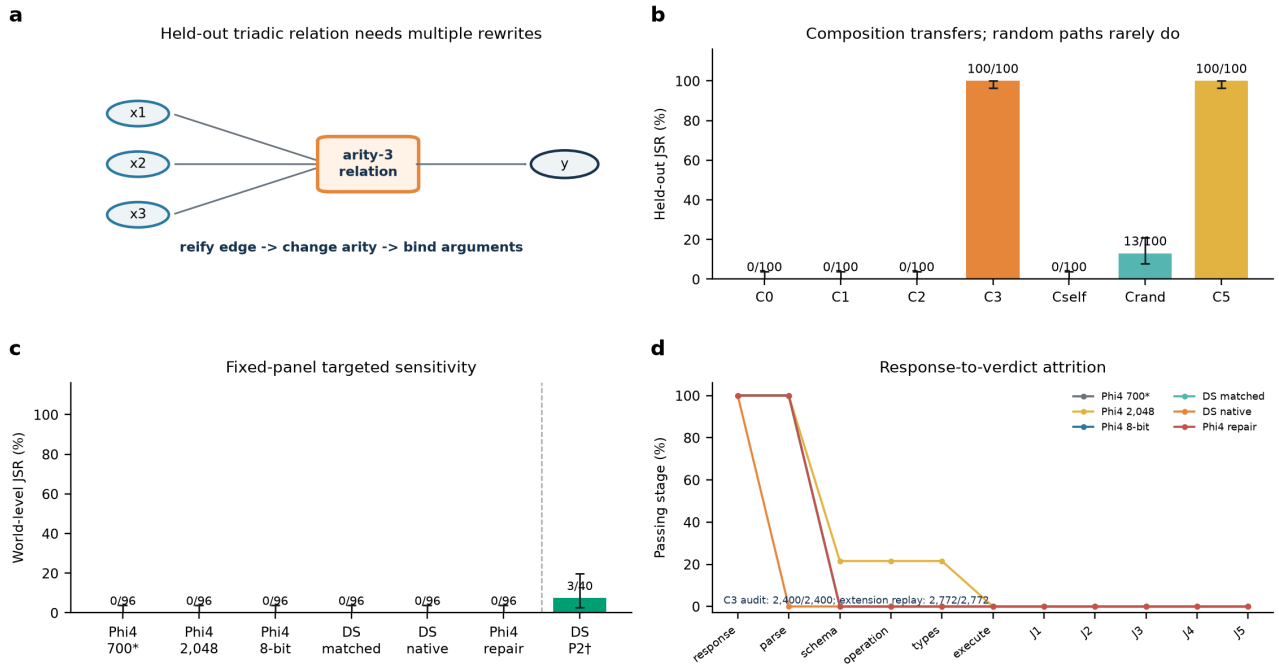


Figure 4 | Held-out transfer, deterministic replay and the minimal targeted sensitivity extension. Panel c reports the fixed 96-world paired panel, except the predeclared $n=40$ supplied-representation control (dagger); the asterisk marks the fixed historical slice of the original $n=400$ Phi-4 confirmatory population. The Phi-4 2,048-token condition changes only the historical completion cap and is not compute-matched. Error bars are Wilson 95% intervals. In panel d, response and structural-validity stages use their explicitly reported response or effective-plan denominators, whereas J1-J5 use executable self-proposed candidates; the connecting lines aid stage ordering but do not imply a common denominator across that boundary. Historical parse validity denotes legacy object extraction, not strict whole-response JSON. No candidate-level significance tests were performed.

We next used a fixed, outcome-blind panel of 96 historical worlds (12 seeds from each of eight families) to test the strongest precision, budget and model-interface alternatives without replacing the original $n=400$ confirmation (Fig. 4). Every autonomous condition remained at 0/96 (Wilson 95% interval 0-3.8%). Relative to the same historical Phi-4 slice, paired JSR differences were 0.000 for Phi-4 4-bit with a 2,048-token cap, Phi-4 8-bit and DeepSeek with the matched interface. DeepSeek native reasoning also differed from its matched condition by 0.000, and one validator-only repair differed from unrepaired Phi-4 8-bit by 0.000. These are targeted sensitivity results, not candidate-level tests or a new confirmatory population.

The attrition pattern remained predominantly pre-executable. On the fixed panel, the longer-budget Phi-4 condition produced schema-valid plans in 993/4,608 registered plan opportunities but no executable plan. In its complete descriptive populations it produced 9/19,200 executable opportunities in the 400 known-family worlds, concentrated in one final candidate that failed J1, and 0/4,800 in 100 held-out worlds; world-level JSR remained 0/400 and 0/100. Phi-4 8-bit produced JSON-extractable responses but 0/4,608 schema-valid plans. The single validator repair left all 4,608 effective opportunities schema-invalid. DeepSeek under the matched interface similarly produced no schema-valid plan (4,480/4,608 opportunities had a non-list plans field and 128 were missing plans). Thus neither increased precision, a longer Phi-4 completion cap, one syntactic repair nor direct substitution of the stronger checkpoint produced an autonomous validated jump under the registered interface.

The native DeepSeek condition also scored 0/96, but for a different interface reason: reasoning text was returned for all 576 calls, while no separate answer reached the parser; 518 calls reached the

4,096-token cap. This does not isolate conceptual proposal quality from reasoning-to-answer serialization. In the balanced supplied-representation control, DeepSeek succeeded in 3/40 worlds (7.5%; Wilson 95% interval 2.6-19.9%). Four of 120 model-authored expression-and-intervention candidates were parse-valid and executable, and three passed J1-J5; 116 calls reached the cap. This positive control establishes that the served checkpoint can sometimes use a supplied correct representation, but not that downstream use is reliable. It also sharpens rather than removes the interface limitation.

A deterministic component audit removes the language model

Inspection of the frozen C3 execution path showed that the representation, expression, candidate ranking and selected intervention were generated deterministically before the model response. The response's representation, expression and intervention fields were overwritten; only its explanation remained, and explanation does not enter J0-J5. We therefore performed a post-hoc, inference-free ablation that replaced both model calls with a valid empty explanation and recomputed every C3 theory and commitment from the archived representation, expression and intervention.

All 2,400 candidate verdicts matched the archive. The model-free replay retained 500/500 jump-world successes and 0/300 control-world false jumps (Fig. 4). Thus Phi-4 was not causally necessary for C3 under the implemented path. AJ5 still measures differences among model-facing proposal interfaces, but the perfect C3 result belongs to the typed deterministic search-and-evaluation scaffold. This audit changes the system attribution, not the validity of the prospective assay.

Transfer to one held-out structural family

The second study reserved triadic_relation_reification from AJ5 development, CJ5 pilot inference and known-family confirmation. Observations supported an incumbent cubic rule under correlated inputs; the target reified an arity-three relation exposed by intervention. C3 succeeded in 100 of 100 held-out worlds, C_rand in 13, and C0, C1, C2 and C_self in none. C3 exceeded C_rand by 0.87 (95% CI 0.80-0.93; adjusted $P=3.00 \times 10^{-4}$).

The holdout is limited. The earlier study contained a binary property-to-relation family, and the generic language already contained reification. What was new was the arity-three structure, product mechanism, generator and instances. Success across a single conceptually adjacent family cannot support broad external generalization.

A worked prospective escape

Figure 5 | One complete prospective escape

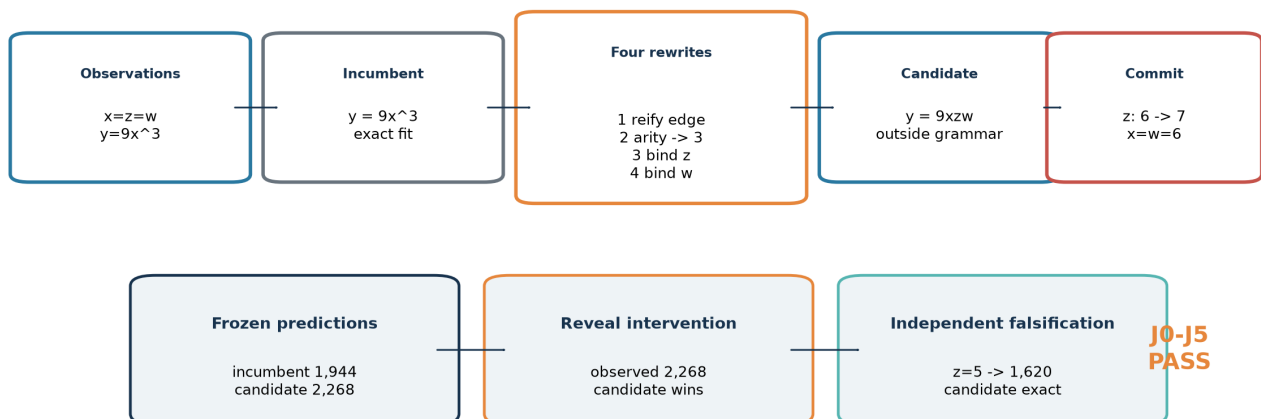


Figure 5 | Worked held-out example. Correlated observations make the cubic incumbent and triadic candidate observationally identical. A committed intervention separates them before outcome reveal, and an independent case falsifies the incumbent.

In one held-out world, observations with $x=z=w$ gave $y=9x^3$, so the incumbent $y=9x^3$ and candidate $y=9xzw$ both fit exactly. Four generic edits reified the incumbent prediction edge, changed the new relation to arity three, and bound z and w (Fig. 5). Before outcome reveal, the assay committed to setting z from 6 to 7 at $x=w=6$. The incumbent predicted 1,944; the candidate predicted 2,268; the revealed outcome was 2,268. Independent falsification at $z=5$ yielded 1,620, again exact for the candidate. The representation failed the frozen language on relation count, type, arity bindings and result structure, and passed J0-J5.

Specificity and replay

The zero false-jump count requires qualification. In C3 controls, 800 of 900 candidates passed J1, 567 passed J2 and 283 passed J3, but none passed J4. The null was therefore non-trivial through the prospective commitment gate. Nevertheless, because no-jump truth is generated inside the incumbent language and the incumbent oracle is exact, an outside candidate cannot strictly beat it under noiseless evaluation. The observed 0/300 is best interpreted as a structural-specificity check under the exact simulator, not an empirical false-positive estimate for noisy science; the Wilson upper bound is descriptive only.

All 10,800 AJ5 and 16,800 CJ5 selected candidates replayed exactly, including 35,533 CJ5 ancestry records. No confirmatory world was excluded and no shard was rerun.

Discussion

This work contributes an assay for a specific event: an executable candidate leaves a frozen hypothesis language and pays for that move with a prospective prediction that beats the best incumbent and survives independent falsification. Canonical non-membership, outcome-before-commitment ordering, proposal-source factorials, false-jump controls and deterministic replay distinguish the assay from broader discovery benchmarks (Table 1).

The experiments also show why component attribution must be audited at code-path level. AJ5 indicates that typed external proposals reach useful representations more often than the registered model interfaces. The targeted sensitivity panel shows that the zero autonomous rate is not obviously explained by Phi-4 4-bit precision, the 700-token cap alone or the identity of the original checkpoint: all paired conditions remained 0/96. However, autonomous candidates still almost never crossed the structural interface, and native DeepSeek reasoning did not emit a separate answer within its frozen budget. These runs therefore do not establish a model-independent conceptual incapacity. C3, meanwhile, did not require model output. The defensible mechanistic result is that typed representation search can cross a frozen space and that proposal-interface attrition can be measured separately from downstream use. Claims of language-model-driven representation change require a path in which model-generated scientific content survives into the committed theory.

External validity is the principal limitation. The worlds are synthetic and noiseless; C3 is saturated; search strata and generators were co-designed; and the operator language embeds graph, function, relation and binding priors. The original confirmation used one model; the extension sampled only one additional checkpoint on a fixed subset, and its native condition was not compute-matched. Only one conceptually adjacent family was held out. World seeds quantify within-generator reliability. The higher-level generalization units are nine structural families, only one of which was held out, so small world-level P values do not substitute for structural breadth.

A decisive extension should freeze the current assay and add independently authored families, distractor primitives, deeper compositions, stochastic observations, partial observability and at least one recognizable mechanistic simulation. On the model side, the next priority is not a broad checkpoint sweep but an interface that reserves separate reasoning and answer budgets and

guarantees only grammar-level serialization without semantic feedback. Broader open and proprietary model comparisons would then be interpretable. These experiments are proposed, not part of the present evidence.

The current conclusion is narrower but useful: hypothesis-space expansion can be measured prospectively and replayed exactly; typed deterministic search can produce it in controlled worlds; and the assay exposes both when an apparent language-model capability is supplied by the surrounding scaffold and when a model comparison is dominated by the proposal interface.

Methods

Study design and protocol freezing

AJ5 and CJ5 protocols were frozen in Git before their reported confirmatory model calls. AJ5 uses commit 895ebb9118ffd0046825b88868621f2a70f69f61. CJ5 uses 65f20874e16bddf8a7ae36996395ff52b27153b7, with a documented correction at 7ecb977 and held-out unlock at 27ee542. The commits are publicly retrievable but unsigned, and no independent registry or transparency-log timestamp was located. We therefore describe the studies as prospectively specified and commit-frozen, not formally preregistered. Pilot seeds were excluded; no confirmatory result changed families, operators, thresholds, budgets or seeds.

Model and inference

Both studies used frozen microsoft/phi-4 revision 2db69c1c3e91a05d2c64a3185acfbaf36f744e25, vLLM 0.10.2, dynamic bitsandbytes 4-bit quantization, a 4,096-token context, temperature 0.2, top-p 0.95 and completion cap 700. AJ5 sampling proposals used temperature 0.7. Each request had a deterministic seed. No fine-tuning or cross-world adaptation occurred.

Minimal targeted sensitivity extension

The original Phi-4 artifacts were preserved at commit ae1ede683fdef09f2bf60f6e1052b60394ad6cf8, tag nmi-phi4-frozen-2026-09 and archive branch nmi-phi4-frozen-archive-2026-09. A separate extension protocol, panel, model configurations and generation code were frozen at commit 320eb29b33ddaed596b0fe7b3f1d5895c706f311 and amended only for operational execution at f846c89287e379fe313551c47765c24f2abf4959. New outputs used the nmi_minimal_sensitivity_v1 namespace and could not overwrite historical files.

An outcome-blind SHA-256 ranking selected 12 of the 50 historical CJ5 seeds before any new model call: 30014, 30012, 30029, 30025, 30011, 30023, 30000, 30032, 30001, 30037, 30002 and 30015. Applying the same seeds to all eight known families produced a fixed 96-world paired panel. The positive control used the first five selected seeds in each family (n=40). This extension is a targeted sensitivity analysis, not a replacement confirmatory population.

The frozen run matrix contained Phi-4 8-bit C_self on 96 worlds; DeepSeek matched C_self with reasoning_effort=none on the same worlds; DeepSeek native C_self with reasoning_effort=max; and a supplied-correct-representation DeepSeek positive control. The DeepSeek service identified the checkpoint as deepseek-ai/DeepSeek-V4-Flash-Vision-Exp, revision 86f746b36186f0e567729a5c06a8c918caba82a9, served as deepseek-v4-flash-vision-exp by vLLM 0.25.2.dev0+g752a3a504.d20260714. We do not equate this checkpoint with any other DeepSeek release. The matched condition retained the historical 700-token output cap; the native and positive-control conditions froze a 4,096-token output cap. Reasoning text was returned separately in message.reasoning; a separate reasoning-token count was recorded only if exposed by the service.

A separately prospectively frozen PHI-BUDGET-SENSITIVITY condition retained the historical Phi-4 revision, 4-bit runtime, prompt semantics, primitive vocabulary, worlds, three candidate slots, 16 plans per slot, J0-J5 gates, interventions and hidden-information boundaries. Its only scientific change was increasing the predeclared completion cap from 700 to 2,048 tokens; it did not add representation

attempts, candidates or interventions and is not compute-matched to the historical condition. The source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. Complete known-family ($n=400$) and held-out ($n=100$) shards were incorporated under outcome-blind amendment NMI-MIN-SENS-V1-AMENDMENT-002 before analysis. The fixed 96-world panel provides the primary paired budget comparison, while the complete populations are reported descriptively.

Worlds, primitive vocabulary, three candidate slots, 16 four-step plans per slot, representation opportunities, J0-J5, prospective interventions and hidden-information boundaries were unchanged. The matched condition received no additional semantic information. The positive control fixed only the correct representation; the model still authored the executable expression, explanation and intervention selection, so success could not be created by overwriting those fields. Because the offline historical cascade had 100% pre-executable attrition, exceeding the frozen 25% trigger, one additional Phi-4 8-bit condition permitted exactly one replacement repair response per structurally invalid slot. Validator feedback named only syntax, schema, operation, reference, type or arity errors and exposed no truth, target distance, outcome or gate value.

Worlds, representations and search

Generators returned public observations, redacted actions, a frozen incumbent LanguageSpec, hidden truth and independent intervention and falsification cases. AJ5 used eight families and 50 jump seeds per family; CJ5 used 50 new seeds per family. No-jump controls used 25 seeds per family in each phase. CJ5 added 100 held-out jump and 100 control seeds. Representations were canonical typed graphs with executable fitted programs.

AJ5 tested direct model output, model sampling, fixed-space reasoning, attribute-only mutation and two typed high-level proposal conditions. CJ5 used 29 generic local rewrites. C3 traversed 48 four-operation branches and retained three candidates by a deterministic score. C_rand sampled 48 four-step paths. C_self requested 16 plans in each of three slots; invalid plans were not repaired.

Prospective evaluation

J0 required incumbent observation $MSE \leq 10^{-12}$; J1 required DSL validity and frozen-language non-membership; J2 required candidate observation $MSE \leq 10^{-12}$; J3 required candidate-oracle prediction separation ≥ 0.5 on the committed action; J4 required intervention MSE improvement > 0.1 ; J5 required falsification $MSE \leq 10^{-12}$ and improvement > 0.1 . Candidate, prediction, action and split hashes were committed before simulator evaluation.

Statistics and component audit

The world was the replicate. AJ5 used 10,000 family-stratified paired bootstrap replicates; CJ5 used the same bootstrap for effects and paired sign-flip tests. Holm correction was applied within prospectively specified comparison families at $\alpha=0.05$. Candidate rows were used only for attrition. Family-level distributions are reported descriptively because eight known families plus one held-out family do not support a stable population-level family inference.

The post-hoc component audit reconstructed all C3 candidates from archived deterministic representations, fitted expressions and selected interventions, replacing the model-derived explanation with an empty string. It then regenerated each world, froze a new commitment and recomputed J0-J5. No model inference was run.

For the targeted sensitivity analysis, exact world counts, JSR and Wilson 95% intervals are primary. Comparisons use paired world transitions and paired JSR differences on the identical 96-world panel; family rates are descriptive. The full $n=400$ known-family and $n=100$ held-out Phi-4 budget populations are supplementary descriptive sensitivities, and the $n=40$ positive control is shown separately. No candidate-level significance tests are performed and P values are not used as primary evidence. Response and plan attrition are reported with their denominators, followed by executable-candidate J1-J5 attrition.

Integrity and replay

Requests, representations, ancestry edges, fitted programs, commitments, gate values and configuration hashes were retained. Replay reconstructed representations and recomputed J0-J5. Infrastructure failures would have required whole-shard reruns; none occurred. There were no outcome-quality exclusions.

AI assistance in research and writing

Phi-4 generated the original registered outputs described above. OpenAI Codex was used after the original confirmatory experiments to inspect artifacts, implement and orchestrate the separately frozen targeted sensitivity extension, recompute inference-free audits, verify literature metadata and assist drafting. It did not choose hypotheses using observed outcomes, alter historical source data or change the frozen evaluation rules. Human authors verified the cited metadata and remain responsible for originality, accuracy and integrity.

Data availability

Synthetic-world definitions, historical and extension result tables, raw call ledgers, manifests and replay artifacts are included in the project repository. The original confirmatory state and separate sensitivity namespace are identified by commits and tags. The submission release will archive an exact commit in a DOI-minting repository. No personal or restricted third-party data were used.

Code availability

Source code for generation, search, evaluation, offline attrition, statistics, figure generation, component audit and replay is included in the project repository. The release will identify the exact commit and archival DOI. Model weights are not redistributed; repository identifier, revision and runtime are reported above.

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Supplementary Information

Supplementary Methods

S1. Formal estimand

Let R_0 denote a world's incumbent typed representation and $H(R_0)$ the frozen set of executable hypotheses admitted by its LanguageSpec. The incumbent oracle h_0^* minimizes observation loss over $H(R_0)$. A candidate representation R is an escape only when it is DSL-valid and $R \notin H(R_0)$ under canonical membership checks. A world-level validated jump requires at least one of three candidates to pass J0-J5; semantic similarity and model judgment are excluded.

The jump success rate is the proportion of jump worlds with a validated candidate. The false-jump rate applies the identical decision to control worlds whose truth is in $H(R_0)$. Abductive precision is the number of validated candidates divided by candidates passing J0-J3. These quantities answer different questions and are never substituted for one another.

S2. Procedural families

AJ5 generated eight families: latent common cause, unification, hidden regimes, property-to-relation, state invention, coordinate transformation, causal ambiguity and meta-law. Each generator returned public observations, an incumbent representation and language, a hidden executable truth, a finite intervention domain and independent falsification cases. Generator validation checked deterministic output, truth redaction, exact incumbent observational fit, existence of a discriminating intervention and reachability of a valid escaped explanation.

Jump seeds were 10000-10049 per family and controls 20000-20024 per family. CJ5 used reconstruction seeds 30000-30049, controls 50000-50024, held-out jump seeds 40000-40099 and held-out controls 60000-60099. World generation used no LLM output.

S3. Typed representation and incumbent membership

A representation is a canonical graph of typed nodes and edges plus executable equations, dependencies, constraints, observability, arity, temporal indices and argument bindings. Canonical serialization sorts unordered fields before hashing. LanguageSpec.contains checks node kinds, edge/relation forms, equation families and other frozen admissibility constraints. J1 records at least one concrete membership failure and cannot be satisfied by renaming or paraphrase.

The incumbent oracle is exhaustive where the admissible grid is finite and bounded by a registered exact procedure otherwise. Any failure of J0 invalidates a world rather than removing it as difficult. No such confirmatory failure occurred.

S4. AJ5 conditions

- B0 direct LLM: directly emits a typed theory graph.
- B1 sample matched: high-temperature independent samples propose at most three typed operations.
- B2 fixed-space agent: representation remains R_0 ; calls may reason and revise inside it.

- B3 attribute mutation: an external mechanism changes only values/equation attributes allowed by $H(R0)$.
- B4 representation mutation: three proposals sampled with replacement from a nine-member typed portfolio.
- B5 full system: three structurally distinct portfolio proposals, archive accounting and deterministic falsification.

The portfolio contained latent, invariant, sign-contrast regime, additive relation, additive state, square-function, affine-context, causal-edge and transition variants. This high-level alignment motivated CJ5 and is not hidden as a limitation.

The P0/P1/P2 factorial reused the same reasoning code and world populations. Only the proposal source differed: LLM, external portfolio or oracle-correct representation. P2 did not receive fitted parameters, truth-program output or intervention outcomes.

S5. CJ5 generic primitive language

GENERIC_PRIMITIVE_SET_V1 contains 29 local operations: add/remove node, edge, function, equation, dependency, constraint and temporal index; reverse/retype/relabel; change observability or arity; bind/unbind; compose/decompose; merge/split; reify edge or node; and bounded copy/crossover. ADD_NODE creates an untyped Primitive; typing, observability, arity, temporal indexing, dependencies and each argument binding require separate records.

Every record binds the parent hash, child hash, operator, canonical arguments, seed and depth. The implementation rejects family names, truth, outcome fields, target distance and family-level aliases such as LATENTIZE, ADD_STATE, PROPERTY_TO_RELATION, ADD_REGIME, COMMON_CAUSE, META_LAW and COORDINATE_TRANSFORM.

S6. CJ5 search and controls

C3 traversed 48 branches of four local edits. Fixed allocations covered generic topologies for typed nodes, function composition, argument-bound functions, relations and reified edges. Ranking used only DSL validity, incumbent non-membership, observational compatibility, existence of an outcome-blind discriminating query, structural novelty and ancestry diversity.

C0 evaluated 192 within-space candidates. C2 evaluated 192 depth-one alternatives. C_rand drew 48 four-step paths and selected final candidates by seed-fixed structural hash. C_self supplied the identical manifest to Phi-4 and requested 16 plans in each of three slots; invalid plans consumed capacity and were not externally completed. C1 retained the AJ5 atomic portfolio as a reference. C5 supplied the target representation and deterministic truth compiler as a conditional ceiling. Because the operation semantics of C1 and C5 differ, they are shown separately in cost comparisons.

S7. Held-out family and lock

triadic_relation_reification uses three correlated observed inputs. The incumbent language fits a cubic rule on the observational support. The target represents an arity-three relation whose product mechanism separates under intervention. A valid construction requires edge reification, an arity change and separate argument bindings. No primitive produces the complete target.

The generator, structure and confirmatory seeds were excluded from AJ5 and CJ5 pilot inference. Only deterministic unit tests for redaction, oracle adequacy and constructive reachability preceded the lock. Known-family and control runs were terminal before the held-out unlock commit. Because AJ5 contained a binary property-to-relation family and the generic set already included reification, this is structural-family hold-out, not wholly concept-free invention.

S8. Candidate fitting and intervention selection

The shared fitter deterministically maps a valid representation and public observations to an executable program. It cannot read hidden truth, validation outcomes or family labels. For J3, the designer evaluates candidate and incumbent predictions over a finite public action set and selects their maximum absolute separation. It records action, predictions, candidate hash and split hash before querying the simulator. Model-proposed actions are retained only as diagnostics.

Thresholds were frozen: J0 and J2 observation $MSE \leq 10^{-12}$; J3 separation ≥ 0.5 ; J4 candidate intervention $MSE < \text{incumbent } MSE - 0.1$; J5 candidate falsification $MSE \leq 10^{-12}$ and $< \text{incumbent } MSE - 0.1$. All six gates are conjunctive.

S9. Opportunity and compute accounting

Every condition-world cell had three final slots, two LLM calls per slot, one final candidate evaluation and one intervention commitment per slot, with 700 completion tokens per call. AJ5 therefore executed 32,400 prospectively specified calls plus 3,600 A6 calls. CJ5 executed 33,600 calls. Unused completion capacity was not reassigned. Actual calls, tokens, attempted/valid operations, candidate evaluations, interventions and latency are reported rather than claiming equality of realized token counts.

S10. Statistical analysis

The replicate was the world. AJ5 used 10,000 deterministic, family-stratified paired bootstrap replicates with seed 20260902. Seeds were resampled inside each family and conditions remained paired; the aggregate gave each family equal weight. Percentile 95% intervals were reported for rates and paired differences. One-sided empirical P values included the finite-replicate correction and eight primary comparisons were Holm adjusted.

CJ5 used the same stratified bootstrap for effect intervals and paired random sign-flip tests for registered beneficial contrasts. C3-C0 and C3-C2 formed the known-family primary multiplicity family; C3-C1, C3-C_rand and C3-C_self formed a secondary family. Held-out contrasts formed their own prospectively specified family. Two-sided Wilson intervals summarize JSR/FJR. No candidate-level pseudoreplication was used.

Post-hoc component audit

An inference-free audit replaced both C3 model calls with a valid empty explanation while preserving only the archived deterministic representation, fitted expression and maximum-separation intervention. Every world was regenerated and every commitment and J0-J5 verdict recomputed. All 2,400 candidate verdicts matched, retaining 500/500 jump successes and 0/300 control false jumps. The audit is explicitly post-hoc and changes attribution, not the registered comparison.

In the 400 known-family worlds, all 1,200 historical C_self responses were non-empty and the registered legacy parser extracted a JSON object, but the outer plans value was never a list. Thus 0/19,200 registered plan opportunities reached execution or structural evaluation. Every response reached the 700-token cap and strict whole-response JSON validity was 0/1,200. Incumbent fallback candidates inserted after empty self-search are excluded from proposal attrition. The analysis made no model calls.

Retained jump gain was $\rho_J = (JSR_{C3} - JSR_{C0}) / (JSR_{C1} - JSR_{C0})$. It is undefined for a non-positive denominator. It compares observed rates and does not normalize unequal operator semantics.

S11. Exclusions, failures and replay

Parse errors, invalid graphs, non-exact fits, absent discriminating interventions and failed falsification were outcomes, not exclusions. A whole shard could be rerun only for a configuration/hash mismatch, unavailable server, missing/duplicate rows or infrastructure interruption, using identical seeds and model configuration. Confirmatory exclusions and shard reruns were both zero.

Replay rebuilt each selected representation from R0 and its ancestry, recomputed canonical hashes and fitted expressions, reconstructed the frozen intervention, and recomputed J0-J5. AJ5 reproduced 10,800/10,800 selected candidates; CJ5 reproduced 16,800/16,800 with 35,533 ancestry records. There were no mismatches.

Minimal targeted sensitivity protocol

Historical Phi-4 results were preserved on a dedicated tag and archive branch. The new protocol used a separate namespace and a commit-frozen panel selected by outcome-blind salted SHA-256 ranking. The same 12 existing seeds were applied to each of eight known families, yielding 96 paired worlds; no new generator or seed was introduced. A balanced supplied-representation positive control used five of those seeds per family (n=40).

The only new model conditions were Phi-4 8-bit C_self, DeepSeek matched C_self (reasoning_effort=None, 700 output tokens), DeepSeek native C_self (reasoning_effort=Max, 4,096 output tokens) and DeepSeek supplied-representation positive control (reasoning_effort=Max, 4,096 output tokens). Offline attrition crossed the prospectively frozen 25% pre-execution trigger, enabling one Phi-4 8-bit repair condition. The repair replaced, rather than augmented, a structurally invalid response and received only validator error classes; it could not receive semantic or simulator feedback.

The separately frozen PHI-BUDGET-SENSITIVITY reused the exact historical Phi-4 revision and 4-bit runtime. Prompt semantics, primitive vocabulary, known and held-out worlds, three candidate slots, 16 four-step plans per slot, representation opportunities, interventions, J0-J5 and all hidden-information boundaries were unchanged. Only generation.max_tokens increased from 700 to 2,048. The condition therefore tests sensitivity to the original completion cap; it is not compute-matched. Its source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. The completed known-family (n=400) and held-out (n=100) shards were incorporated before outcome analysis through amendment NMI-MIN-SENS-V1-AMENDMENT-002; the preselected 96-world panel is the primary paired comparison and the full shards are descriptive.

Each shard had to match its frozen seed panel, model identity, revision, call cardinality and artifact schema before receiving complete_verified status. Replay was locked until all five shards were verified. Analysis was then locked until all outputs replayed with zero mismatches. Reported inference consists of counts, Wilson intervals, paired world transitions, paired JSR differences and per-family descriptions. DeepSeek native is explicitly not compute-matched, and the Phi-4 sensitivity changes serving engine together with precision.

S12. Minimal sensitivity results and attrition

All autonomous fixed-panel conditions produced 0/96 successful worlds (Wilson 95% interval 0-3.8%): the historical Phi-4 4-bit slice, Phi-4 4-bit with the 2,048-token cap, Phi-4 8-bit, DeepSeek matched, DeepSeek native and Phi-4 8-bit with one validator-only repair. The five registered paired contrasts therefore contained 96 joint failures, no successes unique to either member and a paired JSR difference of 0.000. Each of the eight families contributed 12 worlds and had 0/12 in every autonomous condition; these family counts are descriptive.

The fixed-panel 2,048-token Phi-4 run produced 993/4,608 schema-valid plan opportunities but no executable plan. Across the complete descriptive known-family population, 4,642/19,200 opportunities were schema-valid and 9 executable. Those 9 plans occurred within a single selected slot, yielding one

candidate, which failed J1. The held-out population produced 1,289/4,800 schema-valid and 0 executable opportunities. JSR was 0/400 known-family and 0/100 held-out, matching the historical world-level counts.

Phi-4 8-bit produced 4,608/4,608 JSON-extractable effective opportunities but 0 schema-valid because the extracted outer plans field was not a list. Its one-repair condition repeated the same error in every effective replacement opportunity. DeepSeek matched had 4,480 non-list plans errors and 128 missing plans among 4,608 opportunities. DeepSeek native exposed reasoning text in all 576 calls, but no separate answer reached the parser; 518 calls ended at the 4,096-token cap and 58 reported a stop finish reason.

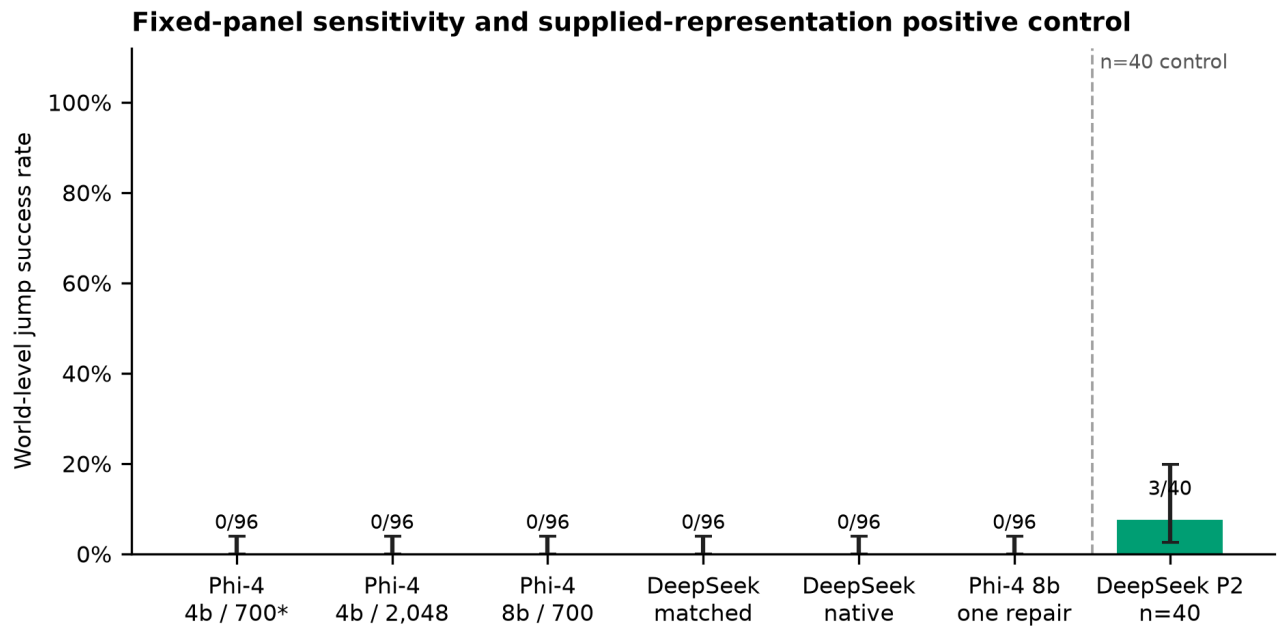
The balanced DeepSeek supplied-representation control succeeded in 3/40 worlds (7.5%; Wilson 95% interval 2.6-19.9%): 1/5 coordinate-transformation and 2/5 hidden-regime worlds, with 0/5 in each other family. Of 120 model-authored expression-and-intervention candidates, 4 were parse-valid, executable and passed J1-J2; 3 also passed J3-J5. The other calls supplied reasoning text without a usable final answer; 116/120 reached the cap. This is a minimum positive control for occasional supplied-representation use, not a reliable ceiling.

The extension used 576 calls in each unrepaired $n=96$ C_self condition, 864 in the triggered repair and 120 in P2. Total completion tokens were 231,773 for Phi-4 8-bit, 237,792 for DeepSeek matched, 2,249,518 for DeepSeek native, 433,373 for repaired Phi-4, 485,312 for P2 and 528,976 for the fixed-panel Phi-budget condition. The service exposed reasoning text but not a separate reasoning-token count for DeepSeek. All five new shards and both Phi-budget populations replayed without mismatch: 2,772 candidate rows, zero mismatches and zero replay-time model calls.

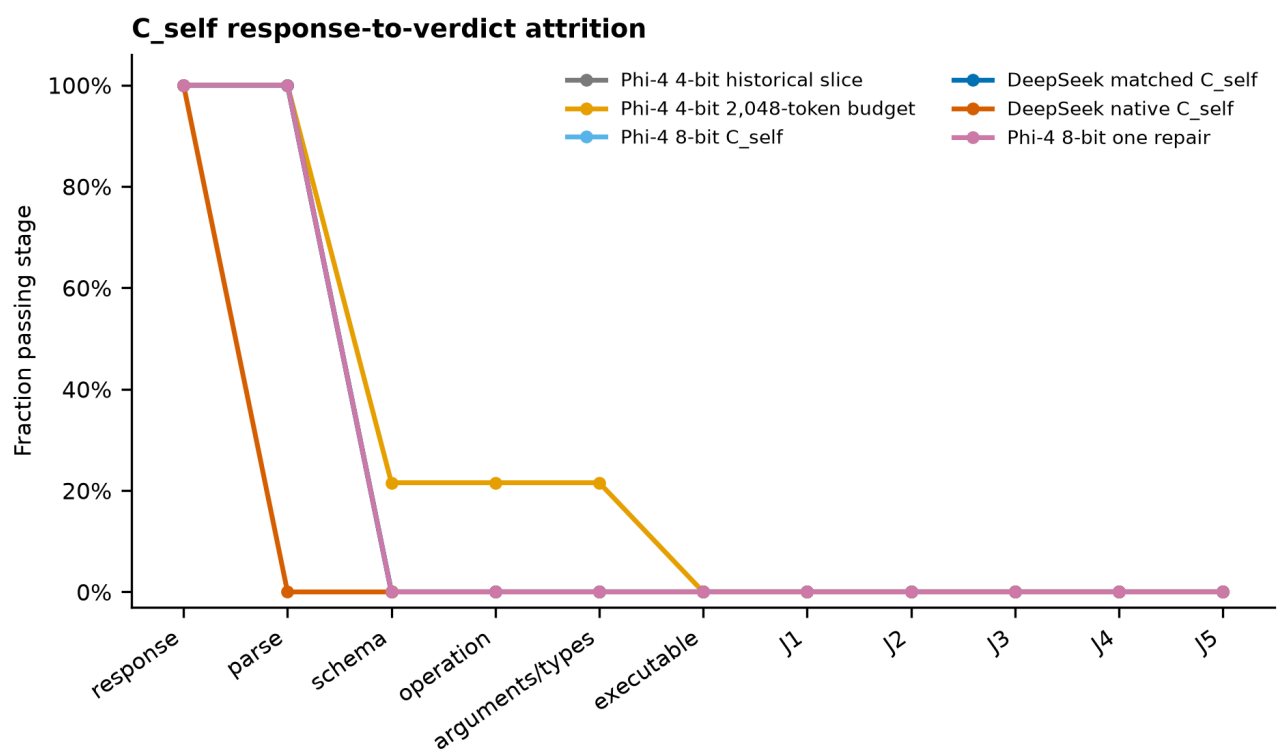
S13. Software and environment

The repository records package versions, configuration hashes, prompt identifiers, model revision, vLLM image digest, GPU class and random seeds in reproducibility manifests. Tests cover world determinism, public-field redaction, DSL membership, primitive legality, budget invariants, reachability, statistics and replay. The publication phase did not call the model or generate new experimental rows.

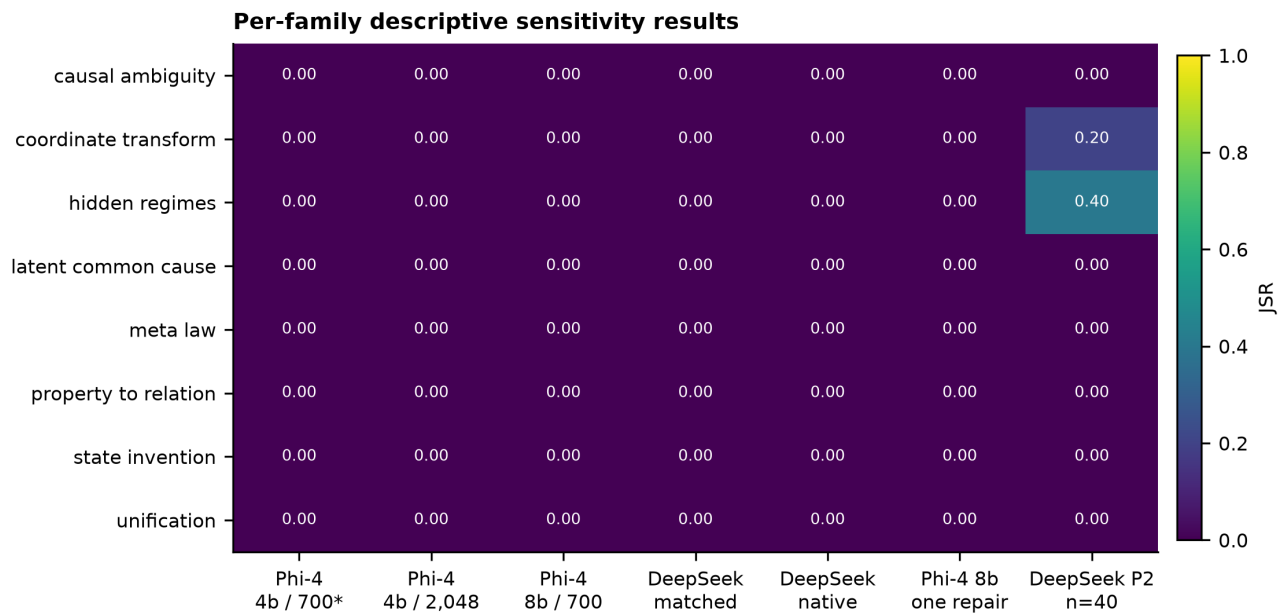
Targeted sensitivity source data



Extended Data Figure 8a | Exact world-level sensitivity results and Wilson 95% intervals. The asterisk identifies the fixed historical slice of the original n=400 confirmation; the supplied-representation control uses a distinct balanced n=40 subset.



Extended Data Figure 8b | Response-to-verdict attrition. Denominators and units are reported in the accompanying source-data tables.



Extended Data Figure 8c | Per-family descriptive sensitivity results; no family-level or candidate-level significance test was performed.

Sensitivity result table

Condition	Population	Success	JSR	Wilson 95% CI
historical phi4 4bit cself	original confirmatory n=400, fixed paired subset shown	0/96	0.0%	0.0-3.8%
phi4 4bit budget cself	previously frozen n=400 budget sensitivity, fixed paired subset shown	0/96	0.0%	0.0-3.8%
phi8 cself	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
deepseek matched cself	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
deepseek native cself	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
phi8 cself repair	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
deepseek p2	new balanced n=40 positive-control subset	3/40	7.5%	2.6-19.9%

Paired world-level transitions

Reference	Comparison	Both fail	Both pass	New only	Old only	JSR difference
historical phi4 4bit cself	phi4 4bit budget cself	96	0	0	0	+0.000
historical phi4 4bit cself	phi8 cself	96	0	0	0	+0.000
historical phi4 4bit cself	deepseek matched cself	96	0	0	0	+0.000
deepseek matched cself	deepseek native cself	96	0	0	0	+0.000
phi8 cself	phi8 cself repair	96	0	0	0	+0.000

Complete cumulative gate attrition

Condition	Unit	Stage	Passed	Denominator	Rate
historical phi4 4bit cself	proposal_response	response_returned	1200	1200	100.0%
historical phi4 4bit cself	proposal_response	parse_valid	1200	1200	100.0%
historical phi4 4bit cself	proposal_response	schema_valid	0	1200	0.0%
historical phi4 4bit cself	proposal_response	operation_valid	0	1200	0.0%
historical phi4 4bit cself	proposal_response	argument_type_valid	0	1200	0.0%
historical phi4 4bit cself	proposal_response	executable	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J1	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J2	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J3	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J4	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J5	0	1200	0.0%
phi8 cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
phi8 cself	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
phi8 cself	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
phi8 cself	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%
phi8 cself	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
phi8 cself	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
phi8 cself	effective_plan_opportunity	executable	0	4608	0.0%
phi8 cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
phi8 cself	executable_self_proposed_candidate	J1	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J2	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J3	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J4	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J5	0	0	0.0%
deepseek matched cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
deepseek matched cself	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
deepseek matched cself	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
deepseek matched cself	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%

Condition	Unit	Stage	Passed	Denominator	Rate
deepseek matched cself	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
deepseek matched cself	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
deepseek matched cself	effective_plan_opportunity	executable	0	4608	0.0%
deepseek matched cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
deepseek matched cself	executable_self_proposed_candidate	J1	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J2	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J3	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J4	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J5	0	0	0.0%
deepseek native cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
deepseek native cself	effective_plan_opportunity	non_empty_answer	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	json_parse_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	executable	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
deepseek native cself	executable_self_proposed_candidate	J1	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J2	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J3	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J4	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J5	0	0	0.0%
phi8 cself repair	effective_plan_opportunity	request_returned	4608	4608	100.0%
phi8 cself repair	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
phi8 cself repair	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
phi8 cself repair	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%

Condition	Unit	Stage	Passed	Denominator	Rate
phi8 cself repair	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
phi8 cself repair	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
phi8 cself repair	effective_plan_opportunity	executable	0	4608	0.0%
phi8 cself repair	effective_plan_opportunity	representation_constructed	0	4608	0.0%
phi8 cself repair	executable_self_proposed_candidate	J1	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J2	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J3	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J4	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J5	0	0	0.0%
phi4 4bit budget cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
phi4 4bit budget cself	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
phi4 4bit budget cself	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
phi4 4bit budget cself	effective_plan_opportunity	plan_schema_valid	993	4608	21.5%
phi4 4bit budget cself	effective_plan_opportunity	operation_names_valid	993	4608	21.5%
phi4 4bit budget cself	effective_plan_opportunity	argument_types_valid	993	4608	21.5%
phi4 4bit budget cself	effective_plan_opportunity	executable	0	4608	0.0%
phi4 4bit budget cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J1	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J2	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J3	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J4	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J5	0	0	0.0%
deepseek p2	model_authored_expression_and_intervention_candidate	parse_valid	4	120	3.3%
deepseek p2	model_authored_expression_and_intervention_candidate	executable	4	120	3.3%
deepseek p2	model_authored_expression_and_intervention_candidate	J1	4	120	3.3%
deepseek p2	model_authored_expression_and_intervention_candidate	J2	4	120	3.3%

Condition	Unit	Stage	Passed	Denominator	Rate
deepseek p2	model_authored_expression_and_intervention_candidate	J3	3	120	2.5%
deepseek p2	model_authored_expression_and_intervention_candidate	J4	3	120	2.5%
deepseek p2	model_authored_expression_and_intervention_candidate	J5	3	120	2.5%

Model-call and compute ledger

Condition	Calls	Attempts	Prompt tokens	Completion tokens	Reasoning text	Latency (s)
phi8 cself	576	576	704376	231773	0	13971.8
deepseek matched cself	576	576	786804	237792	0	10636.3
deepseek native cself	576	576	839796	2249518	576	156547.1
phi8 cself repair	864	864	1254663	433373	0	26462.2
deepseek p2	120	120	183891	485312	120	38836.1
phi4 4bit budget cself	576	576	704376	528976	0	40446.1

Reasoning-token counts are reported only when exposed by the serving API; reasoning-text availability is not converted into an inferred token count.

Full Phi-4 completion-budget sensitivity

Condition	Population	Success	JSR	Wilson 95% CI
historical phi4 4bit cself full	original confirmatory n=400	0/400	0.0%	0.0-1.0%
phi4 4bit budget cself full	budget sensitivity n=400	0/400	0.0%	0.0-1.0%
historical phi4 4bit cself heldout	original held-out n=100	0/100	0.0%	0.0-3.7%
phi4 4bit budget cself heldout	budget sensitivity held-out n=100	0/100	0.0%	0.0-3.7%

Full paired completion-budget transitions

Reference	Comparison	Both fail	Both pass	New only	Old only	Difference
historical phi4 4bit cself full	phi4 4bit budget cself full	400	0	0	0	+0.000
historical phi4 4bit cself heldout	phi4 4bit budget cself heldout	100	0	0	0	+0.000

Full Phi-4 completion-budget gate attrition

Population	Unit	Stage	Passed	Denominator	Rate
phi4 4bit budget cself full	effective_plan_opportunity	request_returned	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	non_empty_answer	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	json_parse_valid	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	plan_schema_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	operation_names_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	argument_types_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	executable	9	19200	0.0%
phi4 4bit budget cself full	effective_plan_opportunity	representation_constructed	9	19200	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J1	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J2	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J3	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J4	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J5	0	1	0.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	request_returned	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	non_empty_answer	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	json_parse_valid	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	plan_schema_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	operation_names_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	argument_types_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	executable	0	4800	0.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	representation_constructed	0	4800	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J1	0	0	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J2	0	0	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J3	0	0	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J4	0	0	0.0%

Population	Unit	Stage	Passed	Denominator	Rate
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J5	0	0	0.0%

Full Phi-4 completion-budget compute ledger

Population	Calls	Attempts	Prompt tokens	Completion tokens	Latency (s)
phi4 4bit budget cself full	2400	2400	2921915	2216914	169361.1
phi4 4bit budget cself heldout	600	600	696489	546374	41524.5

Protocol and claim audit

NMI Claim Matrix

Directly supported

- In eight procedural world families, direct LLM proposal and matched additional sampling each produced 1/400 validated jumps, while fixed-space search and value-only mutation produced 0/400.
- External representation-level mutation produced 142/400 validated jumps in both B4 and B5 and succeeded in every tested family; B5 did not outperform B4.
- With the reasoning path held fixed, LLM, external and oracle representation proposals produced 0/400, 142/400 and 400/400 successes.
- A 29-operation generic rewrite language contained constructive solutions at bounded depths 2-4, while none of 17,280 depth-one alternatives validated.
- Registered structured composition produced 400/400 known-family and 100/100 held-out successes. Matched depth-one search produced 0/400 and 0/100; matched random composition produced 52/400 and 13/100.
- The frozen Phi-4 self-composition condition produced 0/400 known and 0/100 held-out successes under the same primitive vocabulary and registered opportunity budget.
- The atomic high-level reference produced 131/400 known-family successes but 0/100 on the held-out structural family.
- C3 exceeded C_rand by 87 percentage points in both populations; both registered Holm-adjusted P values were 0.00029997.
- No false jumps were observed for any condition in 200 known-family and 100 held-out no-jump worlds. This does not imply zero population risk.
- All 10,800 AJ5 and 16,800 CJ5 selected candidates replayed deterministically with zero mismatches.
- Offline reconstruction of the original known-family Phi-4 C_self interface found 1,200/1,200 non-empty responses but 0/1,200 strict whole-response JSON parses and 0/19,200 executable plan opportunities; all responses reached the registered 700-token completion cap. This is an interface-attrition result, not evidence about the quality of valid proposals.
- All five minimal-extension shards were completion-verified, and inference-free replay reproduced 2,772/2,772 candidate rows with zero mismatches and zero model calls.
- On the identical fixed 96-world panel, historical Phi-4 4-bit, Phi-4 4-bit with a 2,048-token cap, Phi-4 8-bit, DeepSeek matched, DeepSeek native and Phi-4 8-bit with one validator repair each produced 0/96 world successes (Wilson 95% interval 0-3.8%); every registered paired JSR difference was 0.000.
- The separately frozen Phi-4 completion-budget condition retained the historical revision, 4-bit runtime, prompts, worlds, slots, attempts, interventions and J0-J5, changing only the cap from 700 to 2,048 tokens. It produced 0/400 known-family and 0/100 held-out successes. In the known-family population, 4,642/19,200 plan opportunities were schema-valid and 9 were executable, but these yielded one final candidate that failed J1; held-out executability was 0/4,800.
- Phi-4 8-bit, DeepSeek matched and the one-repair Phi-4 condition produced 0/4,608 schema-valid effective plan opportunities on the fixed panel. DeepSeek native returned reasoning text for all 576 calls but no separate parseable answer, with 518/576 calls reaching the 4,096-token cap.

- The balanced DeepSeek supplied-representation control produced 3/40 successful worlds (7.5%; Wilson 95% interval 2.6-19.9%). Four of 120 model-authored expression-and-intervention candidates were parse-valid and executable, and three passed J1-J5; 116/120 calls reached the cap.

The n=96 panel is a targeted sensitivity analysis and must never be described as replacing the original n=400 confirmatory study. The fixed n=96 slice is the primary paired Phi-budget comparison; the complete known-family n=400 and held-out n=100 budget populations are supplementary descriptive sensitivities, not new confirmatory populations. The P2 n=40 control is balanced but not paired to every n=96 condition and must be shown separately.

Interpretation, not directly proven

- Representation proposal and search organization may be a major bottleneck for the tested frozen model, but the original C_self result alone cannot separate that possibility from truncation and serialization failure.
- The unchanged 0/96 rates show that 4-bit precision, the historical 700-token cap alone, one validator-only repair and simple substitution of the tested stronger checkpoint do not by themselves remove the bottleneck under these interfaces.
- Because autonomous candidates almost never reached execution, the extension still cannot distinguish weak conceptual proposals from structural serialization failure. Native DeepSeek's 0/96 is especially an interface result because reasoning consumed the frozen output budget without a separate answer.
- DeepSeek P2 shows that the served checkpoint can sometimes use a supplied representation; 3/40 is a limited positive control, not evidence of reliable supplied-representation reasoning.
- Abductive reasoning may be usefully decomposed into representation proposal, downstream reasoning and prospective validation.
- A phenomenologically discontinuous explanatory jump may sometimes be implemented as a sequence of computationally local representation transformations.
- Structured search priors, rather than raw sampling volume alone, appear responsible for much of C3's advantage over random composition.
- The held-out result suggests limited structural generalization beyond family-aligned atomic operators, not concept invention from scratch.

Any sentence expressing these points must use interpretive qualifiers and remain tied to the controlled procedural setting.

Prohibited claims

- LLMs generally cannot jump or are incapable of abduction.
- Human creativity, intuition or scientific creativity has been reproduced or explained.
- The system autonomously invented representations from nothing.
- The results establish general scientific discovery, AGI or arbitrary theory discovery.
- Programmatic representation mutation is equivalent to human intuition.
- Zero observed false jumps means the true false-jump rate is zero.
- World models or agents are unnecessary.
- The representation meta-language is unconstrained.
- The result applies to all models, domains or LLMs.

- The targeted $n=96$ sensitivity panel is a new preregistered or replacement confirmatory population.
- The held-out family is wholly conceptually novel.
- The study proves Zahavy's position wrong or proves a computational impossibility claim.

Maximum publication claim

In controlled procedural worlds, structured search composed supplied generic representation rewrites into prospectively validated explanations outside locally adequate incumbent hypothesis spaces, including an unseen structural configuration withheld from operator development. The result does not establish autonomous representation invention or general scientific discovery.